PHYS 143 – Mechanics and Thermal Physics

Vectors

Reference and practice deck

Supporting material cut from the live lecture. Nothing here is new physics.

Hope Donglo

Office 24, Department of Physics, University of Ghana

How to use this deck

Labels

REFERENCE keep for later use: definitions, equation anatomy, toolkits, derived results.

PRACTICE use to develop the skill: concept checks, extra examples, independent problems.

Where things are

Part A Contact 1: vocabulary, triangle toolkits, sine and cosine rules.
Part B Contact 2: component toolkits, extra examples.
Part C Contact 3: product anatomy, toolkits, extra examples.
Part D the chain of ideas and the seven toolkits.

The live lecture is `lecture01_vectors.pdf`; the master toolkit there is the one procedure to use first.

What to master, and what supports it

Core skills you should master

- distinguish scalars from vectors;
- add and subtract vectors graphically;
- resolve vectors into components;
- determine component signs;
- determine the correct quadrant;
- reconstruct magnitude and direction;
- add vectors using components;
- use unit-vector notation and calculate the unit vector of a vector;
- calculate and interpret a dot product;
- calculate and interpret a cross product.

Supporting tools

- detailed sine-rule and cosine-rule treatment;
- more elaborate vector-triangle cases;
- extended 3D material (supplement).

Supporting slides carry the label *supporting tool* in their titles. They stay in the deck for reference; the core skills are what the problems test first.

Distance and displacement

Distance

scalar
depends on the total path travelled

Displacement

vector
runs directly from initial position to final position

Question

You walk once around a $400\,\mathrm{m}$ track and stop where you started. What is your displacement?

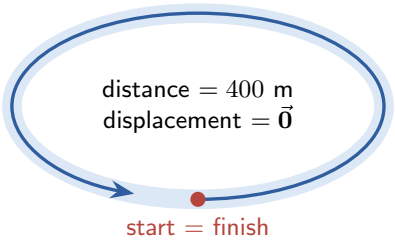
Distance and displacement

Distance

scalar
depends on the total path travelled

Displacement

vector
runs directly from initial position to final position



Question

You walk once around a $400\,\mathrm{m}$ track and stop where you started. What is your displacement?

Zero, although the distance travelled is $400\,\mathrm{m}$.

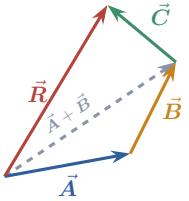
But what if... displacement is a vector. How do we name, draw and compare vectors? → **vocabulary**.

Defining the vocabulary of vectors

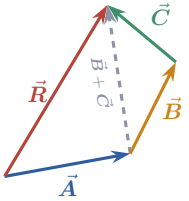
Term	What it is	What it measures	Units	Representation
Magnitude	the size of a vector; a scalar, never negative	how much	those of the vector (m, m/s, N)	length of the arrow
Direction	the orientation of a vector in space	which way	degrees, or compass words	orientation of the arrow; angle θ
Displacement	straight line from the initial to the final position	how far, and which way, you are from the start	m	arrow from start to finish
Resultant	the single vector equal to a sum of vectors	the net effect of several vectors	those of the vectors added	arrow from first tail to last head

Interpret: magnitude answers *how much?*, direction answers *which way?*; a vector answers both. Component and unit vector are defined in Contact 2.

Three or more vectors: associativity



$(\vec{A}+\vec{B})+\vec{C}$



$\vec{A}+(\vec{B}+\vec{C})$

Associative law

$(\vec{A}+\vec{B})+\vec{C}=\vec{A}+(\vec{B}+\vec{C})$. The vectors may be grouped in any way; the resultant is the same.

But what if... we want to undo an addition, or find the change between two vectors? → **subtraction**.

Concept check: before any numbers

Answer each with yes or no, and justify it with a sketch

1. Can a vector have a negative magnitude?
2. Can the displacement be zero while the distance travelled is not?
3. Can two vectors have the same magnitude but different directions? Are they equal?
4. Can $|\vec{A} + \vec{B}|$ be smaller than both $|\vec{A}|$ and $|\vec{B}|$?

Answer each with yes or no, and justify it with a sketch

1. Can a vector have a negative magnitude?
2. Can the displacement be zero while the distance travelled is not?
3. Can two vectors have the same magnitude but different directions? Are they equal?
4. Can $|\vec{A} + \vec{B}|$ be smaller than both $|\vec{A}|$ and $|\vec{B}|$?

Answers

1. **No.** $|\vec{A}| \geq 0$; a minus sign in $-\vec{A}$ reverses the direction, it does not make the magnitude negative.
2. **Yes.** One lap of a track: displacement $\vec{0}$, distance $400\,\mathrm{m}$.
3. **Yes, and they are not equal.** Equal vectors need the same magnitude *and* direction.
4. **Yes.** $5\,\mathrm{m}$ east plus $4\,\mathrm{m}$ west gives $1\,\mathrm{m}$ east.

TOOLKIT 1 – Directions and angles

WHEN TO USE: a direction is given in words, or must be reported as an angle.

PROCEDURE: $\text{E} = 0^\circ$, $\text{N} = 90^\circ$, $\text{W} = 180^\circ$, $\text{S} = 270^\circ$.
"30° north of east": start east, turn north. "30° east of north": start north, turn east.
Standard θ is measured counter-clockwise from $+x$. Always sketch before converting.

CHECK: the sketch and the angle θ must point the same way.

Predict first

Convert to θ: 30° east of north, 40° south of west, 20° west of south.

TOOLKIT 1 – Directions and angles

WHEN TO USE: a direction is given in words, or must be reported as an angle.

PROCEDURE: E $= 0°$, N $= 90°$, W $= 180°$, S $= 270°$.
"$30°$ north of east": start east, turn north. "$30°$ east of north": start north, turn east.
Standard θ is measured counter-clockwise from $+x$. Always sketch before converting.

CHECK: the sketch and the angle θ must point the same way.

Predict first

Convert to θ: $30°$ east of north, $40°$ south of west, $20°$ west of south.

$\theta = 60°$, $\theta = 220°$, $\theta = 250°$.

Adding two vectors at 90°

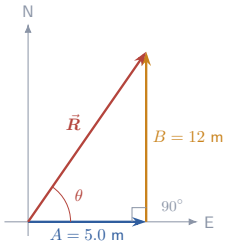
Predict first

You walk $5.0\,\mathrm{m}$ east, then $12\,\mathrm{m}$ north. Is the displacement $17\,\mathrm{m}$? Predict its size and direction before calculating.

Adding two vectors at 90°

Predict first

You walk $5.0\,\mathrm{m}$ east, then $12\,\mathrm{m}$ north. Is the displacement $17\,\mathrm{m}$? Predict its size and direction before calculating.



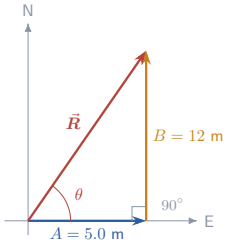
Head to tail, the two legs meet at a right angle, so $\vec{A}$, $\vec{B}$ and $\vec{R}$ form a **right triangle**, and $\vec{R}$ is its longest side (the hypotenuse).

But what if... we know R lies between 12 and $17\,\mathrm{m}$, but exactly how long is it? Why would the hypotenuse follow a simple rule? → **Pythagoras**.

Adding two vectors at 90°

Predict first

You walk $5.0\,\mathrm{m}$ east, then $12\,\mathrm{m}$ north. Is the displacement $17\,\mathrm{m}$? Predict its size and direction before calculating.



Head to tail, the two legs meet at a right angle, so $\vec{A}$, $\vec{B}$ and $\vec{R}$ form a **right triangle**, and $\vec{R}$ is its longest side (the hypotenuse).

Prediction: $12 < R < 17\,\mathrm{m}$ (a straight line is shorter than the two legs), leaning toward north.

But what if... we know R lies between 12 and $17\,\mathrm{m}$, but exactly how long is it? Why would the hypotenuse follow a simple rule? → **Pythagoras**.

TOOLKIT 2 – Vector triangles (right-angle case)

WHEN TO USE: two vectors are added head to tail and the legs meet at $90°$.

PROCEDURE: **1.** Construct the head-to-tail triangle; $\vec{\boldsymbol{R}}$ runs from the first tail to the last head. **2.** Check the angle between the legs. **3.** $R = \sqrt{A^2 + B^2}$, $\theta = \tan^{-1}(B/A)$.

CHECK: $|A - B| \leq R \leq A + B$; for 5.0 and $12\,\mathrm{m}$, $7.0 \leq R \leq 17\,\mathrm{m}$, and $13\,\mathrm{m}$ passes.

WATCH OUT: the internal angle is not always the angle you were given. Next: when it is not $90°$.

Predict first

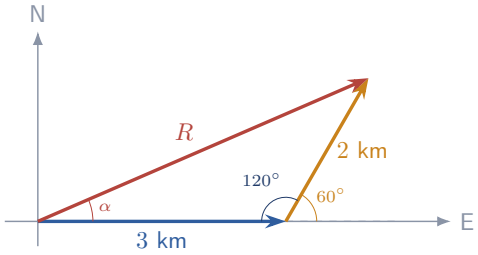
For the opening problem (1 km north, 2 km east) what is the internal angle? Which tool applies?

Your turn

A cyclist rides **3.00 km east**, then **2.00 km at 60° north of east**. Find the resultant displacement.

Predict first

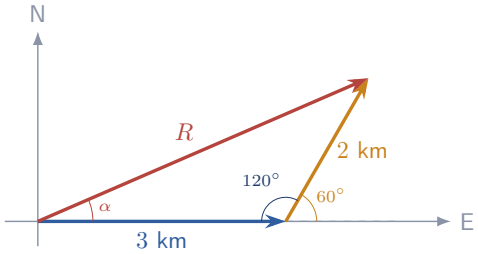
Sketch it. Is Pythagoras still allowed? Is R closer to 5 km or to 3 km?



Limitation of Pythagoras: when the angle is not 90°

Your turn

A cyclist rides **3.00 km east**, then **2.00 km at 60° north of east**. Find the resultant displacement.



Predict first

Sketch it. Is Pythagoras still allowed? Is R closer to 5 km or to 3 km?

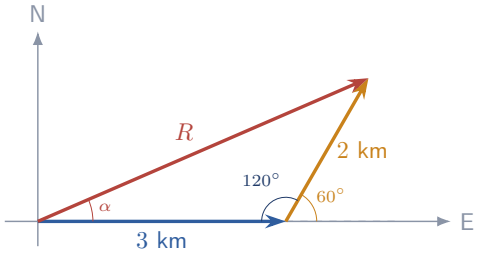
The triangle has no right angle: the angle *inside* the triangle between the two legs is $180^\circ - 60^\circ = 120^\circ$, not the 60° given. Pythagoras does not apply directly.

But what if... we need a tool for *any* triangle? → the **cosine and sine rules**.

Limitation of Pythagoras: when the angle is not 90°

Your turn

A cyclist rides **3.00 km east**, then **2.00 km at 60° north of east**. Find the resultant displacement.

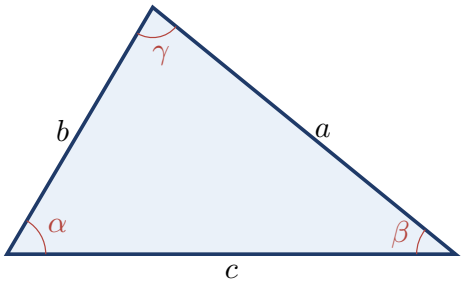


Predict first

Sketch it. Is Pythagoras still allowed? Is R closer to 5 km or to 3 km?

The triangle has no right angle: the angle *inside* the triangle between the two legs is $180^\circ - 60^\circ = 120^\circ$, not the 60° given. Pythagoras does not apply directly. **Proof of failure:** Pythagoras gives $3.61\,\text{km}$, but the second leg alone carries you $2.00\cos 60^\circ = 1.00\,\text{km}$ further east: at least $4.00\,\text{km}$ from the start. So $3.61\,\text{km}$ is impossible.

But what if... we need a tool for *any* triangle? → the **cosine and sine rules**.



side a is opposite angle α, etc.

Cosine rule: find a side

$$a^2 = b^2 + c^2 - 2bc\cos\alpha$$

For two vectors, $R^2 = A^2 + B^2 - 2AB\cos\gamma$, with γ the internal angle opposite $\vec{\boldsymbol{R}}$.

Sine rule: find an angle

$$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma}$$

These are mathematical tools, not a new vector idea; we use them without derivation. If $\alpha = 90^\circ$ the cosine rule becomes Pythagoras.

TOOLKIT 2 – Vector triangles (completed)

WHEN TO USE: two vectors are added head to tail at any angle between the legs.

PROCEDURE: **1.** Construct head to tail. **2.** Find the internal angle γ opposite $\vec{R}$. **3.** Right angle: Pythagoras. Otherwise the cosine rule for a side, $R^2 = A^2 + B^2 - 2AB\cos\gamma$, then the sine rule for an angle.

CHECK: $|A - B| \leq R \leq A + B$. For $3\,\mathrm{m}$ and $4\,\mathrm{m}$, $1\,\mathrm{m} \leq R \leq 7\,\mathrm{m}$, so a calculated $9\,\mathrm{m}$ must be wrong.

WATCH OUT: γ is the internal angle of the triangle. It is not always the angle you were given.

Which angle is which? Triangle angle and vector-product angle

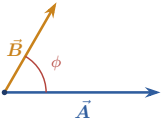
Predict first

In the cyclist problem the given angle is $60°$. What is the internal angle γ opposite R?

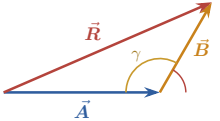
Which angle is which? Triangle angle and vector-product angle

Predict first

In the cyclist problem the given angle is $60°$. What is the internal angle γ opposite R?



tail to tail: $\phi = 60°$



head to tail: $\gamma = 120°$

Rule: two different angles

γ: internal angle of the head-to-tail triangle ($120°$, so the cosine rule uses $\cos 120°$).
ϕ: tail-to-tail angle between the vectors, used later in dot and cross products ($60°$ here).
The triangle angle γ and the vector-product angle ϕ are not necessarily the same angle.

Cyclist example: solution (supporting tool)

Set up: sketch; the 120° angle is opposite R. Predict $3 < R < 5$ km.

Execute, cosine rule:

$$R^2 = 3.00^2 + 2.00^2 - 2(3.00)(2.00)\cos 120^\circ = 19.0$$

$$R = 4.36\,\text{km}$$

Cyclist example: solution (supporting tool)

Set up: sketch; the 120° angle is opposite R. Predict $3 < R < 5$ km.

Execute, cosine rule:

$$R^2 = 3.00^2 + 2.00^2 - 2(3.00)(2.00)\cos 120^\circ = 19.0$$

$$R = 4.36\,\text{km}$$

sine rule for the direction α at the start:

$$\frac{\sin\alpha}{2.00} = \frac{\sin 120^\circ}{4.36} \;\Rightarrow\; \alpha = 23.4^\circ$$

$\vec{\boldsymbol{R}} = 4.36\,\text{km}$ at 23.4° north of east.

Cyclist example: solution (supporting tool)

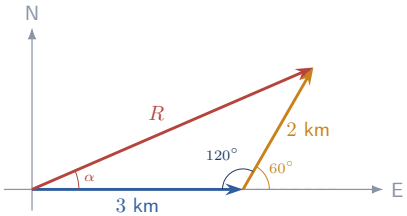
Set up: sketch; the 120° angle is opposite R. Predict $3 < R < 5$ km.
Execute, cosine rule:

$$R^2 = 3.00^2 + 2.00^2 - 2(3.00)(2.00)\cos 120^\circ = 19.0$$

$$R = 4.36\,\text{km}$$

sine rule for the direction α at the start:

$$\frac{\sin\alpha}{2.00} = \frac{\sin 120^\circ}{4.36} \;\Rightarrow\; \alpha = 23.4^\circ$$



$\vec{\boldsymbol{R}} = 4.36\,\text{km}$ at 23.4° north of east. **Evaluate:** units km; $3 < R < 5$ as predicted; $0^\circ < \alpha < 60^\circ$, matching the sketch. ✓

But what if... the rider makes four legs? Each leg needs another triangle, with its own cosine and sine rule. Is there a systematic method? → **components**.

EQUATION – components

Meaning. A_x, A_y give how much of $\vec{\boldsymbol{A}}$ lies along each axis.

Variables. A magnitude; θ angle from $+x$.

Units. A_x, A_y carry the units of $\vec{\boldsymbol{A}}$; θ is in degrees.

Interpretation. the sign says right or left, up or down; $|A_x| \leq A$.

Predict first

For $\theta = 150°$, is A_x positive or negative? Is A_y?

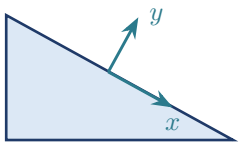
But what if... the vector points left or down, as at $150°$? Then $\cos\theta$ or $\sin\theta$ is negative. → **quadrants**.

TOOLKIT 3 – Components

WHEN TO USE: a vector is given by a magnitude and an angle, and you need its parts along chosen axes.

PROCEDURE: **Choose axes** (perpendicular, not necessarily horizontal and vertical). **Sketch and predict signs:** right or left gives the sign of A_x; up or down gives the sign of A_y. **Resolve:** $A_x = A\cos\theta$, $A_y = A\sin\theta$ for θ from $+x$; otherwise *adjacent* $\rightarrow$ *cosine, opposite* $\rightarrow$ *sine*, with signs from the sketch.

CHECK: the signs match the sketch and $\sqrt{A_x^2 + A_y^2} = A$. A_x is a number; $A_x\hat{\boldsymbol{\imath}}$ is the component vector.



axes tilted with a slope

Decide, then compare with your neighbour

1. Can a component be negative? Can it be zero for a non-zero vector?
2. Can $|A_x|$ be larger than the magnitude A?
3. If you rotate the axes, does the vector change? Do its components change?

Decide, then compare with your neighbour

1. Can a component be negative? Can it be zero for a non-zero vector?
2. Can $|A_x|$ be larger than the magnitude A?
3. If you rotate the axes, does the vector change? Do its components change?

Answers

1. **Yes to both.** The sign gives the direction along the axis; a vector along y has $A_x = 0$.
2. **No.** $A_x = A\cos\theta$ and $|\cos\theta| \leq 1$.
3. **The vector does not change; its components do.** Components describe the vector *relative to chosen axes*, which is why we choose axes to make a problem simple.

Sketch, predict the signs, then calculate

Take $+x$ east, $+y$ north. $\vec{\boldsymbol{D}}_1$: $10.0\,\mathrm{m}$ south-west. $\vec{\boldsymbol{D}}_2$: $12.0\,\mathrm{m}$, 30° east of south. Name the quadrant and the signs of the components of each, then find them.

Predict first

Name the axis each angle is measured from before choosing $\sin$ or $\cos$.

Sketch, predict the signs, then calculate

Take $+x$ east, $+y$ north. $\vec{\boldsymbol{D}}_1$: $10.0\,\mathrm{m}$ south-west. $\vec{\boldsymbol{D}}_2$: $12.0\,\mathrm{m}$, 30° east of south. Name the quadrant and the signs of the components of each, then find them.

Predict first

Name the axis each angle is measured from before choosing $\sin$ or $\cos$.

$\vec{\boldsymbol{D}}_1$**:** quadrant III, both components negative; south-west is 45° from west, so $D_{1x} = -10.0\cos 45^\circ = -7.07\,\mathrm{m}$ and $D_{1y} = -7.07\,\mathrm{m}$.

$\vec{\boldsymbol{D}}_2$**:** quadrant IV, $D_{2x} > 0$ and $D_{2y} < 0$; the 30° is measured from the south $(-y)$ axis, so $D_{2x} = +12.0\sin 30^\circ = +6.00\,\mathrm{m}$ and $D_{2y} = -12.0\cos 30^\circ = -10.4\,\mathrm{m}$.

But what if... the angle is measured from the y axis? Then $\cos$ goes with y and $\sin$ with x: the function follows the axis the angle touches.

Finding direction from components

TOOLKIT 4 – Reconstructing a vector

WHEN TO USE: the components are known and you need the magnitude and direction.

PROCEDURE: $A = \sqrt{A_x^2 + A_y^2}$. **1.** Quadrant from the signs of A_x, A_y. **2.** Reference angle $\theta_{\text{ref}} = \tan^{-1}|A_y/A_x|$. **3.** Place it: I: θ_{ref}; II: $180^\circ - \theta_{\text{ref}}$; III: $180^\circ + \theta_{\text{ref}}$; IV: $360^\circ - \theta_{\text{ref}}$.

CHECK: the direction agrees with the sketch.

WATCH OUT: vectors with the same A_y/A_x can point in different directions. **Quadrant first, calculation second.**

$\vec{P} = (-2, 2)$: left and up, quadrant II.
$\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$

$$\theta_P = 180^\circ - 45^\circ = \mathbf{135}^\circ$$

TOOLKIT 4 – Reconstructing a vector

WHEN TO USE: the components are known and you need the magnitude and direction.

PROCEDURE: $A = \sqrt{A_x^2 + A_y^2}$. **1.** Quadrant from the signs of A_x, A_y. **2.** Reference angle $\theta_{\text{ref}} = \tan^{-1}|A_y/A_x|$. **3.** Place it: I: θ_{ref}; II: $180^\circ - \theta_{\text{ref}}$; III: $180^\circ + \theta_{\text{ref}}$; IV: $360^\circ - \theta_{\text{ref}}$.

CHECK: the direction agrees with the sketch.

WATCH OUT: vectors with the same A_y/A_x can point in different directions. **Quadrant first, calculation second.**

$\vec{P} = (-2, 2)$: left and up, quadrant II.
$\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$

$$\theta_P = 180^\circ - 45^\circ = \mathbf{135}^\circ$$

$\vec{Q} = (2, -2)$: right and down, quadrant IV.
$\theta_{\text{ref}} = 45^\circ$

$$\theta_Q = 360^\circ - 45^\circ = \mathbf{315}^\circ \ (= -45^\circ)$$

TOOLKIT 4 – Reconstructing a vector

WHEN TO USE: the components are known and you need the magnitude and direction.

PROCEDURE: $A = \sqrt{A_x^2 + A_y^2}$. **1.** Quadrant from the signs of A_x, A_y. **2.** Reference angle $\theta_{\text{ref}} = \tan^{-1}|A_y/A_x|$. **3.** Place it: I: θ_{ref}; II: $180^\circ - \theta_{\text{ref}}$; III: $180^\circ + \theta_{\text{ref}}$; IV: $360^\circ - \theta_{\text{ref}}$.

CHECK: the direction agrees with the sketch.

WATCH OUT: vectors with the same A_y/A_x can point in different directions. **Quadrant first, calculation second.**

$\vec{P} = (-2,\ 2)$: left and up, quadrant II.
$\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$

$$\theta_P = 180^\circ - 45^\circ = \mathbf{135}^\circ$$

$\vec{Q} = (2,\ -2)$: right and down, quadrant IV.
$\theta_{\text{ref}} = 45^\circ$

$$\theta_Q = 360^\circ - 45^\circ = \mathbf{315}^\circ\ (= -45^\circ)$$

Your turn: $A_x = -3.0\,\text{m}$, $A_y = -4.0\,\text{m}$. Find A and θ.

But what if... there are several vectors? How do their components combine? → **adding vectors**.

TOOLKIT 4 – Reconstructing a vector

WHEN TO USE: the components are known and you need the magnitude and direction.

PROCEDURE: $A = \sqrt{A_x^2 + A_y^2}$. **1.** Quadrant from the signs of A_x, A_y. **2.** Reference angle $\theta_{\text{ref}} = \tan^{-1}|A_y/A_x|$. **3.** Place it: I: θ_{ref}; II: $180^\circ - \theta_{\text{ref}}$; III: $180^\circ + \theta_{\text{ref}}$; IV: $360^\circ - \theta_{\text{ref}}$.

CHECK: the direction agrees with the sketch.

WATCH OUT: vectors with the same A_y/A_x can point in different directions. **Quadrant first, calculation second.**

$\vec{P} = (-2, 2)$: left and up, quadrant II.
$\theta_{\text{ref}} = \tan^{-1}(2/2) = 45^\circ$

$$\theta_P = 180^\circ - 45^\circ = \mathbf{135}^\circ$$

$\vec{Q} = (2, -2)$: right and down, quadrant IV.
$\theta_{\text{ref}} = 45^\circ$

$$\theta_Q = 360^\circ - 45^\circ = \mathbf{315}^\circ \ (= -45^\circ)$$

Your turn: $A_x = -3.0\,\text{m}$, $A_y = -4.0\,\text{m}$. Find A and θ. $A = 5.0\,\text{m}$; quadrant III;
$\theta_{\text{ref}} = \tan^{-1}(4/3) = 53.1^\circ$;
$\theta = 180^\circ + 53.1^\circ = 233.1^\circ$ ✓

But what if... there are several vectors? How do their components combine? → **adding vectors**.

Adding vectors with components

Question

If $\vec{\boldsymbol{R}} = \vec{\boldsymbol{A}} + \vec{\boldsymbol{B}} + \vec{\boldsymbol{C}} + \cdots$, how are the components of $\vec{\boldsymbol{R}}$ related to theirs?

Adding vectors with components

Question

If $\vec{\boldsymbol{R}} = \vec{\boldsymbol{A}} + \vec{\boldsymbol{B}} + \vec{\boldsymbol{C}} + \cdots$, how are the components of $\vec{\boldsymbol{R}}$ related to theirs?

Rule: adding vectors by components

$$R_x = \sum A_x, \qquad R_y = \sum A_y \qquad \text{(in 3D also } R_z = \sum A_z\text{)}$$

TOOLKIT 5 – Adding vectors by components

WHEN TO USE: two or more vectors must be added, whatever their directions.
PROCEDURE: **RESOLVE** each vector (a table helps), **COMBINE** like components, **RECONSTRUCT** the magnitude and direction of $\vec{\boldsymbol{R}}$ (Toolkit 4). **Sketch → predict signs → calculate.**
CHECK: $\vec{\boldsymbol{R}}$ lies in the quadrant predicted by the sketch, and R is no larger than the sum of the magnitudes.

But what if... the vectors are given as magnitudes and compass directions? → **multiple vectors**.

Stating the direction in words

A calculation may give $\theta = 129°$, but the physical answer is often clearer as a compass direction. Measure from the nearest cardinal axis:

$$\text{north} = 90°, \qquad 129° - 90° = 39°$$

so $\theta = 129°$ is $39°$ **west of north**. Check by drawing it.

Your turn

Express $\theta = 233.1°$ and $\theta = 96.2°$ as compass directions.

Stating the direction in words

A calculation may give $\theta = 129°$, but the physical answer is often clearer as a compass direction. Measure from the nearest cardinal axis:

$$\text{north} = 90°, \qquad 129° - 90° = 39°$$

so $\theta = 129°$ is $39°$ **west of north**. Check by drawing it.

Your turn

Express $\theta = 233.1°$ and $\theta = 96.2°$ as compass directions.

$233.1° - 180° = 53.1°$ south of west;
$96.2° - 90° = 6.2°$ west of north.

Practice problem

A hiker walks $4.00\,\mathrm{km}$ at 30.0° north of east, then $6.00\,\mathrm{km}$ at 40.0° west of north, then $3.00\,\mathrm{km}$ south. Find the magnitude and direction of the hiker's resultant displacement.

Predict first

Sketch first. Which quadrant will $\vec{\boldsymbol{R}}$ be in?

Practice problem

A hiker walks $4.00\,\text{km}$ at 30.0° north of east, then $6.00\,\text{km}$ at 40.0° west of north, then $3.00\,\text{km}$ south. Find the magnitude and direction of the hiker's resultant displacement.

Predict first

Sketch first. Which quadrant will $\vec{\boldsymbol{R}}$ be in?

θ: 30.0°, 130.0°, 270°
$R_x = 3.46 - 3.86 + 0 = -0.39\,\text{km}$
$R_y = 2.00 + 4.60 - 3.00 = 3.60\,\text{km}$

Practice problem

A hiker walks $4.00\,\text{km}$ at 30.0° north of east, then $6.00\,\text{km}$ at 40.0° west of north, then $3.00\,\text{km}$ south. Find the magnitude and direction of the hiker's resultant displacement.

Predict first

Sketch first. Which quadrant will $\vec{\boldsymbol{R}}$ be in?

θ: 30.0°, 130.0°, 270°
$R_x = 3.46 - 3.86 + 0 = -0.39\,\text{km}$
$R_y = 2.00 + 4.60 - 3.00 = 3.60\,\text{km}$

$R = 3.62\,\text{km}$
left and up: quadrant II
$\theta_{\text{ref}} = \tan^{-1}(3.60/0.39) = 83.8^\circ$
$\theta = 180^\circ - 83.8^\circ = 96.2^\circ$, i.e. 6.2° west of north.

But what if... writing components in words is long? $\rightarrow$ **unit vectors**.

Adding vectors in unit-vector form

Rule: adding in unit-vector form

Collect like terms:

$$\vec{\boldsymbol{A}} + \vec{\boldsymbol{B}} = (A_x + B_x)\hat{\boldsymbol{\imath}} + (A_y + B_y)\hat{\boldsymbol{\jmath}} + (A_z + B_z)\hat{\boldsymbol{k}}$$

Predict first

$\vec{\boldsymbol{A}} = (2\hat{\boldsymbol{\imath}} + 3\hat{\boldsymbol{\jmath}})\,\mathrm{m}$ and $\vec{\boldsymbol{B}} = (\hat{\boldsymbol{\imath}} - 5\hat{\boldsymbol{\jmath}})\,\mathrm{m}$. Is $\vec{\boldsymbol{A}} + \vec{\boldsymbol{B}}$ up or down, left or right of the origin?

Adding vectors in unit-vector form

Rule: adding in unit-vector form

Collect like terms:

$$\vec{\boldsymbol{A}} + \vec{\boldsymbol{B}} = (A_x + B_x)\hat{\boldsymbol{\imath}} + (A_y + B_y)\hat{\boldsymbol{\jmath}} + (A_z + B_z)\hat{\boldsymbol{k}}$$

Predict first

$\vec{\boldsymbol{A}} = (2\hat{\boldsymbol{\imath}} + 3\hat{\boldsymbol{\jmath}})\,\mathrm{m}$ and $\vec{\boldsymbol{B}} = (\hat{\boldsymbol{\imath}} - 5\hat{\boldsymbol{\jmath}})\,\mathrm{m}$. Is $\vec{\boldsymbol{A}} + \vec{\boldsymbol{B}}$ up or down, left or right of the origin?

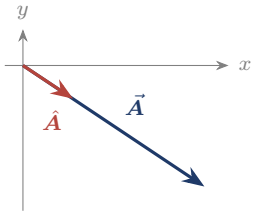
$\vec{\boldsymbol{A}} + \vec{\boldsymbol{B}} = (3\hat{\boldsymbol{\imath}} - 2\hat{\boldsymbol{\jmath}})\,\mathrm{m}$: right and down, quadrant IV. $|\vec{\boldsymbol{A}} + \vec{\boldsymbol{B}}| = \sqrt{3^2 + (-2)^2} = 3.6\,\mathrm{m}$. ✓

Unit vector along any vector

Rule: the unit vector of a vector

With $A = |\vec{A}|$:

$$\hat{A} = \frac{\vec{A}}{A} = \frac{A_x}{A}\,\hat{\imath} + \frac{A_y}{A}\,\hat{\jmath}\,\left(+\tfrac{A_z}{A}\,\hat{k}\right)$$



Divide every component by the magnitude. The result has magnitude 1, no units, and the same direction as $\vec{A}$: dividing by a positive scalar changes the length, not the direction.

Predict first

For $\vec{A} = (3.00\hat{\imath} - 2.00\hat{\jmath})\,\mathrm{m}$: are the components of $\hat{A}$ bigger or smaller than 1? Which signs? Which quadrant?

But what if... does the rule give the right numbers? → **a worked example**.

A vector from its unit vector

Rule: a vector from its unit vector

$\vec{\boldsymbol{A}} = A\,\hat{\boldsymbol{A}}$: magnitude (with units) times direction (pure number).

Your turn

$\vec{\boldsymbol{B}} = (-4.00\hat{\boldsymbol{\imath}} + 7.00\hat{\boldsymbol{\jmath}})\,\mathrm{m}$. Find B and $\hat{\boldsymbol{B}}$, and check $|\hat{\boldsymbol{B}}| = 1$. Then write the force $\vec{\boldsymbol{F}}$ of magnitude $50.0\,\mathrm{N}$ along $\hat{\boldsymbol{B}}$.

A vector from its unit vector

Rule: a vector from its unit vector

$\vec{\boldsymbol{A}} = A\,\hat{\boldsymbol{A}}$: magnitude (with units) times direction (pure number).

Your turn

$\vec{\boldsymbol{B}} = (-4.00\hat{\boldsymbol{\imath}} + 7.00\hat{\boldsymbol{\jmath}})\,\mathrm{m}$. Find B and $\hat{\boldsymbol{B}}$, and check $|\hat{\boldsymbol{B}}| = 1$. Then write the force $\vec{\boldsymbol{F}}$ of magnitude $50.0\,\mathrm{N}$ along $\hat{\boldsymbol{B}}$.

$B = \sqrt{65}\,\mathrm{m} = 8.06\,\mathrm{m}$
$\hat{\boldsymbol{B}} = -0.496\hat{\boldsymbol{\imath}} + 0.868\hat{\boldsymbol{\jmath}}$
check: $\frac{16}{65} + \frac{49}{65} = 1$ ✓

A vector from its unit vector

Rule: a vector from its unit vector

$\vec{\boldsymbol{A}} = A\,\hat{\boldsymbol{A}}$: magnitude (with units) times direction (pure number).

Your turn

$\vec{\boldsymbol{B}} = (-4.00\hat{\boldsymbol{\imath}} + 7.00\hat{\boldsymbol{\jmath}})\,\mathrm{m}$. Find B and $\hat{\boldsymbol{B}}$, and check $|\hat{\boldsymbol{B}}| = 1$. Then write the force $\vec{\boldsymbol{F}}$ of magnitude $50.0\,\mathrm{N}$ along $\hat{\boldsymbol{B}}$.

$B = \sqrt{65}\,\mathrm{m} = 8.06\,\mathrm{m}$
$\hat{\boldsymbol{B}} = -0.496\hat{\boldsymbol{\imath}} + 0.868\hat{\boldsymbol{\jmath}}$
check: $\frac{16}{65} + \frac{49}{65} = 1$ ✓

$\vec{\boldsymbol{F}} = 50.0\,\mathrm{N}\,\hat{\boldsymbol{B}}$
$= (-24.8\hat{\boldsymbol{\imath}} + 43.4\hat{\boldsymbol{\jmath}})\,\mathrm{N}$

But what if... all examples so far are planar? Real forces are three-dimensional. → **3D vectors**.

The 3D Cartesian system

Rule: right-handed axes

Curl the fingers of the right hand from $+x$ to $+y$; the thumb points along $+z$.

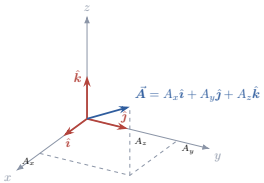
Rule: components and magnitude in 3D

$$\vec{\boldsymbol{A}} = A_x\hat{\boldsymbol{\imath}} + A_y\hat{\boldsymbol{\jmath}} + A_z\hat{\boldsymbol{k}}$$

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Physical meaning

(Extended material, see the supplement.) The mutually perpendicular unit vectors $\hat{\boldsymbol{\imath}}$, $\hat{\boldsymbol{\jmath}}$ and $\hat{\boldsymbol{k}}$ provide three independent directions, so a vector is uniquely specified by its x-, y- and z-components.



But what if... we can only add and subtract? How do we multiply vectors? → **Contact 3**.

The angle ϕ: tail to tail

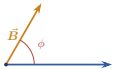
Predict first

The cyclist's legs are $3.00\,\mathrm{km}$ east and $2.00\,\mathrm{km}$ at 60° north of east. Is the angle ϕ for a dot or cross product 60° or 120°?

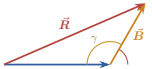
The angle ϕ: tail to tail

Predict first

The cyclist's legs are $3.00\,\mathrm{km}$ east and $2.00\,\mathrm{km}$ at 60° north of east. Is the angle ϕ for a dot or cross product 60° or 120°?



tail to tail: $\phi = 60^\circ$



head to tail: $\gamma = 120^\circ$

Rule: the angle ϕ

For dot and cross products, ϕ is the angle between the vectors when placed tail-to-tail. Here $\phi = 60^\circ$.

Common error

Do not use the internal angle of a head-to-tail vector triangle as ϕ. That angle is $\gamma = 120^\circ$.

EQUATION – dot product

Meaning. $\vec{A} \cdot \vec{B}$ measures how much $\vec{A}$ points along $\vec{B}$, scaled by the size of $\vec{B}$.

Variables. A, B are the magnitudes; ϕ is the angle between the vectors tail to tail, $0^\circ \le \phi \le 180^\circ$; $A_x, \ldots, B_z$ are components.

Units. the product of the units: $\mathrm{m} \times \mathrm{m} = \mathrm{m}^2$; for force and displacement, $\mathrm{N\,m} = \mathrm{J}$.

Interpretation. positive: the vectors broadly agree in direction; zero: perpendicular; negative: they broadly oppose. The result is a **scalar**.

Example. the next slide, then the pair $\vec{A} = \hat{\imath} + 9\hat{\jmath} + 2\hat{k}$, $\vec{B} = -5\hat{\imath} + \hat{\jmath} - \hat{k}$.

Two forms, one scalar

$$\vec{A} \cdot \vec{B} = AB\cos\phi$$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Predict first

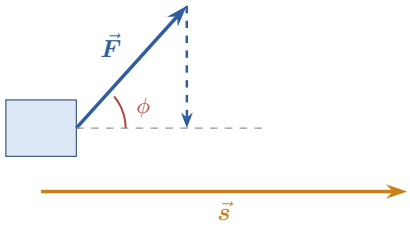
Which form is easier when you have a diagram and two angles? Which when you have components?

TOOLKIT 6 – Dot product

WHEN TO USE: you need a scalar: how much of one vector lies along another, the angle between two vectors, or work.

PROCEDURE: Two forms, one quantity: $\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}} = AB\cos\phi = A_xB_x + A_yB_y + A_zB_z$. Predict the sign: acute angle $\rightarrow$ positive; perpendicular $\rightarrow$ zero; obtuse $\rightarrow$ negative. Angle: $\cos\phi = \dfrac{\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}}}{AB}$, with A, B the magnitudes. Preview: work, $W = \vec{\boldsymbol{F}} \cdot \vec{\boldsymbol{s}}$.

CHECK: the sign agrees with the geometry (positive $\rightarrow \phi < 90^\circ$, negative $\rightarrow \phi > 90^\circ$), and ϕ is the tail-to-tail angle, not γ.

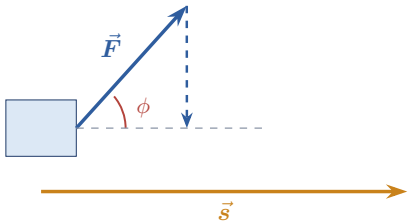


Practice example: work

A rope pulls a crate with $F = 30\,\mathrm{N}$ at $\phi = 40°$ above the horizontal while the crate moves $s = 5.0\,\mathrm{m}$. Find the work done by the rope.

Predict first

Positive or negative? Larger or smaller than $Fs = 150\,\mathrm{J}$?



Practice example: work

A rope pulls a crate with $F = 30\,\mathrm{N}$ at $\phi = 40^\circ$ above the horizontal while the crate moves $s = 5.0\,\mathrm{m}$. Find the work done by the rope.

Predict first

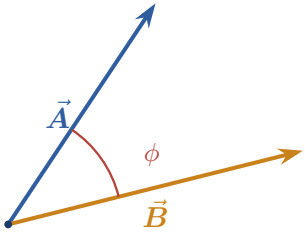
Positive or negative? Larger or smaller than $Fs = 150\,\mathrm{J}$?

Identify: $W = \vec{F} \cdot \vec{s}$. **Set up:** ϕ is the angle between $\vec{F}$ and $\vec{s}$ (tail to tail). **Execute:** $W = Fs\cos\phi = (30)(5.0)\cos 40^\circ = 115\,\mathrm{J}$. **Evaluate:** positive and below $150\,\mathrm{J}$, as predicted; $\mathrm{N\,m} = \mathrm{J}$. ✓

Perpendicular force: $\phi = 90^\circ$ gives $W = 0$ (carrying a bag at constant height).

But what if... what is the angle between the two vectors, when they are given by components rather than a picture? → **angle between vectors**.

Angle between two vectors



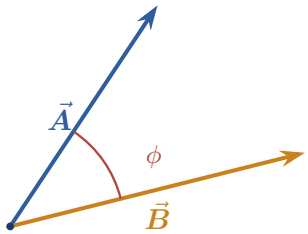
Practice example: angle between vectors

$\vec{A} = 2\hat{\imath} + 3\hat{\jmath}$ and $\vec{B} = 4\hat{\imath} + \hat{\jmath}$. Find the angle ϕ between them.

Predict first

Both lie in quadrant I. Is ϕ acute or obtuse? Is $\vec{A} \cdot \vec{B}$ positive?

Angle between two vectors



Practice example: angle between vectors

$\vec{A} = 2\hat{\imath} + 3\hat{\jmath}$ and $\vec{B} = 4\hat{\imath} + \hat{\jmath}$. Find the angle ϕ between them.

Predict first

Both lie in quadrant I. Is ϕ acute or obtuse? Is $\vec{A} \cdot \vec{B}$ positive?

Execute: $\vec{A} \cdot \vec{B} = (2)(4) + (3)(1) = 11, \quad A = \sqrt{13} = 3.61, \quad B = \sqrt{17} = 4.12$

$$\cos\phi = \frac{11}{(3.61)(4.12)} = 0.740 \;\Rightarrow\; \phi = 42.3^\circ$$

Evaluate: positive dot product, acute angle, as predicted. Check with directions: $\tan^{-1}(3/2) - \tan^{-1}(1/4) = 56.3^\circ - 14.0^\circ = 42.3^\circ$. ✓

One vector pair for the rest of the week

We will use these throughout Contact 3

$$\vec{\boldsymbol{A}} = \hat{\boldsymbol{\imath}} + 9\hat{\boldsymbol{\jmath}} + 2\hat{\boldsymbol{k}}, \qquad \vec{\boldsymbol{B}} = -5\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - \hat{\boldsymbol{k}}$$

Your turn: find $\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}}$ and the angle ϕ between $\vec{\boldsymbol{A}}$ and $\vec{\boldsymbol{B}}$.

One vector pair for the rest of the week

We will use these throughout Contact 3

$$\vec{\boldsymbol{A}} = \hat{\boldsymbol{\imath}} + 9\hat{\boldsymbol{\jmath}} + 2\hat{\boldsymbol{k}}, \qquad \vec{\boldsymbol{B}} = -5\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - \hat{\boldsymbol{k}}$$

Your turn: find $\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}}$ and the angle ϕ between $\vec{\boldsymbol{A}}$ and $\vec{\boldsymbol{B}}$.

$$\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}} = (1)(-5) + (9)(1) + (2)(-1) = 2$$

One vector pair for the rest of the week

We will use these throughout Contact 3

$$\vec{\boldsymbol{A}} = \hat{\boldsymbol{\imath}} + 9\hat{\boldsymbol{\jmath}} + 2\hat{\boldsymbol{k}}, \qquad \vec{\boldsymbol{B}} = -5\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - \hat{\boldsymbol{k}}$$

Your turn: find $\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}}$ and the angle ϕ between $\vec{\boldsymbol{A}}$ and $\vec{\boldsymbol{B}}$.

$$\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}} = (1)(-5) + (9)(1) + (2)(-1) = 2$$

$$A = \sqrt{1 + 81 + 4} = \sqrt{86} = 9.27, \qquad B = \sqrt{25 + 1 + 1} = \sqrt{27} = 5.20$$

$$\cos\phi = \frac{\vec{\boldsymbol{A}} \cdot \vec{\boldsymbol{B}}}{AB} = \frac{2}{(9.27)(5.20)} = 0.0415 \;\Rightarrow\; \phi = 87.6^\circ$$

Evaluate: small positive dot product $\Rightarrow$ angle just under 90°. ✓

But what if... a push perpendicular to a door's lever arm gives $\vec{\boldsymbol{r}} \cdot \vec{\boldsymbol{F}} = 0$, yet turns it most? $\rightarrow$ **cross product**.

Rule: unit-vector cross products

Cyclic order $\hat{\imath} \to \hat{\jmath} \to \hat{k} \to \hat{\imath}$ gives $+$: $\hat{\imath} \times \hat{\jmath} = \hat{k}$, $\hat{\jmath} \times \hat{k} = \hat{\imath}$, $\hat{k} \times \hat{\imath} = \hat{\jmath}$.
Reversed order gives $-$: $\hat{\jmath} \times \hat{\imath} = -\hat{k}$, $\hat{k} \times \hat{\jmath} = -\hat{\imath}$, $\hat{\imath} \times \hat{k} = -\hat{\jmath}$.
Also $\hat{\imath} \times \hat{\imath} = \hat{\jmath} \times \hat{\jmath} = \hat{k} \times \hat{k} = \vec{0}$.

$\times$	$\hat{\imath}$	$\hat{\jmath}$	$\hat{k}$
$\hat{\imath}$	$\vec{0}$	$\hat{k}$	$-\hat{\jmath}$
$\hat{\jmath}$	$-\hat{k}$	$\vec{0}$	$\hat{\imath}$
$\hat{k}$	$\hat{\jmath}$	$-\hat{\imath}$	$\vec{0}$

row $\times$ column

EQUATION – cross product

Meaning. $\vec{A} \times \vec{B}$ is a vector perpendicular to both; its magnitude is largest when the vectors are perpendicular and zero when they are parallel.

Variables. A, B are the magnitudes; ϕ is the angle between the vectors tail to tail, $0^\circ \leq \phi \leq 180^\circ$.

Units. the product of the units: $\mathrm{m} \times \mathrm{m} = \mathrm{m}^2$; for position and force, $\mathrm{N\,m}$.

Interpretation. the magnitude is the area of the parallelogram on the two vectors; the direction is given by the right-hand rule; the order matters.

Example. the worked example on 30° vectors, then the pair $\vec{A}$, $\vec{B}$ used for the dot product.

Magnitude and direction

$$|\vec{A} \times \vec{B}| = AB \sin\phi$$

Direction: right-hand rule.

Predict first

When is $|\vec{A} \times \vec{B}|$ largest?
When is it zero?

Cross product from components

Rule: cross product from components

$$\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = (A_y B_z - A_z B_y)\,\hat{\boldsymbol{\imath}} + (A_z B_x - A_x B_z)\,\hat{\boldsymbol{\jmath}} + (A_x B_y - A_y B_x)\,\hat{\boldsymbol{k}}$$

Memory aid: the determinant

$$\vec{\boldsymbol{A}} \times \vec{\boldsymbol{B}} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

The $\hat{\boldsymbol{\jmath}}$ term takes a minus sign when the determinant is expanded.

Connect: torque, Lecture 8

$$\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{F}}$$

A force $\vec{\boldsymbol{F}}$ applied at position $\vec{\boldsymbol{r}}$ from a pivot. Torque is studied fully in Lecture 8.

But what if... we want a physical application? A force applied at a distance from a pivot. → **torque**.

TOOLKIT 7 – Cross product

WHEN TO USE: you need a vector perpendicular to two vectors, such as torque.

PROCEDURE: **Magnitude:** $|\vec{A} \times \vec{B}| = AB \sin\phi$.
Direction: right-hand rule; $\vec{A} \times \vec{B} = -(\vec{B} \times \vec{A})$.
Unit vectors: cyclic $\hat{\imath} \to \hat{\jmath} \to \hat{k} \to \hat{\imath}$ gives $+$; reversed gives $-$; $\hat{\imath} \times \hat{\imath} = \vec{0}$.
Preview: torque, $\vec{\tau} = \vec{r} \times \vec{F}$.

Components:

$$(A_y B_z - A_z B_y)\,\hat{\imath}$$
$$+(A_z B_x - A_x B_z)\,\hat{\jmath}$$
$$+(A_x B_y - A_y B_x)\,\hat{k}$$

CHECK: $(\vec{A} \times \vec{B}) \cdot \vec{A} = 0$ and $(\vec{A} \times \vec{B}) \cdot \vec{B} = 0$; parallel $\to \vec{0}$ $(\sin 0^\circ = \sin 180^\circ = 0)$; perpendicular $\to$ maximum magnitude.

Concept check: the two products

Decide before calculating

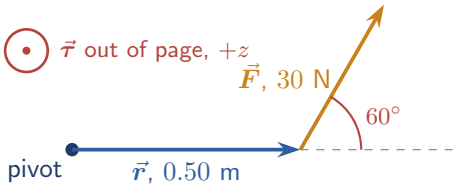
1. Is $\vec{A} \cdot \vec{B}$ a scalar or a vector? Can it be negative?
2. Can $\vec{A} \cdot \vec{B} = 0$ for two non-zero vectors? Can $\vec{A} \times \vec{B} = \vec{0}$?
3. Is $\vec{A} \times \vec{B}$ ever parallel to $\vec{A}$?

Decide before calculating

1. Is $\vec{A} \cdot \vec{B}$ a scalar or a vector? Can it be negative?
2. Can $\vec{A} \cdot \vec{B} = 0$ for two non-zero vectors? Can $\vec{A} \times \vec{B} = \vec{0}$?
3. Is $\vec{A} \times \vec{B}$ ever parallel to $\vec{A}$?

Answers

1. **Scalar.** Yes, when $\phi > 90°$.
2. **Yes to both.** Perpendicular vectors give $\vec{A} \cdot \vec{B} = 0$; parallel or antiparallel vectors give $\vec{A} \times \vec{B} = \vec{0}$.
3. **Never.** $\vec{A} \times \vec{B}$ is perpendicular to both $\vec{A}$ and $\vec{B}$ (check: its dot product with each is zero).

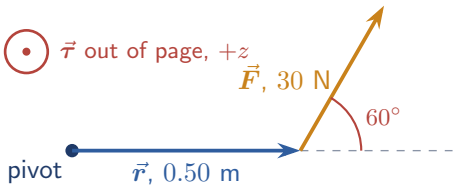


Practice example: torque

A wrench is pivoted at one end. A force $F = 30\,\mathrm{N}$ acts at $r = 0.50\,\mathrm{m}$ from the pivot, at $60°$ to $\vec{r}$ (in the page). Find $\vec{\tau} = \vec{r} \times \vec{F}$.

Predict first

Is τ larger or smaller than $rF = 15\,\mathrm{N\,m}$?
Which way does $\vec{\tau}$ point?



Practice example: torque

A wrench is pivoted at one end. A force $F = 30\,\mathrm{N}$ acts at $r = 0.50\,\mathrm{m}$ from the pivot, at $60°$ to $\vec{r}$ (in the page). Find $\vec{\tau} = \vec{r} \times \vec{F}$.

Predict first

Is τ larger or smaller than $rF = 15\,\mathrm{N\,m}$?
Which way does $\vec{\tau}$ point?

Execute: $\tau = rF\sin\phi = (0.50)(30)\sin 60° = 13.0\,\mathrm{N\,m}$. Right hand: fingers along $\vec{r}$, curled toward $\vec{F}$, thumb out of the page: $+z$.
Evaluate: $13.0 < 15$, as predicted; $\tau = 0$ if $\vec{F} \parallel \vec{r}$ and $\tau = rF$ if $\vec{F} \perp \vec{r}$. ✓

But what if... a problem uses several of these ideas at once: adding, resolving, dot and cross products? Which tool, and when? → **integrated applications**.

Independent application

A rope force of $40.0\,\mathrm{N}$ acts at 60.0° north of east. The object moves $2.00\,\mathrm{m}$ north, then $2.00\,\mathrm{m}$ west.
(a) Find $\vec{s}$. (b) Find the work done by the force. (c) The force acts at $\vec{r} = 0.500\hat{\imath}\,\mathrm{m}$ from a pivot; find $\vec{\tau}$.
Verify (b) with $Fs\cos\phi$ and (c) with $rF\sin\phi$ and the right-hand rule.

Predict first

Sketch first. Which quadrant is $\vec{s}$ in? Is ϕ less than 90°?

Independent application

A rope force of $40.0\,\mathrm{N}$ acts at 60.0° north of east. The object moves $2.00\,\mathrm{m}$ north, then $2.00\,\mathrm{m}$ west.
(a) Find $\vec{s}$. (b) Find the work done by the force. (c) The force acts at $\vec{r} = 0.500\hat{\imath}\,\mathrm{m}$ from a pivot; find $\vec{\tau}$.
Verify (b) with $Fs\cos\phi$ and (c) with $rF\sin\phi$ and the right-hand rule.

Predict first

Sketch first. Which quadrant is $\vec{s}$ in? Is ϕ less than 90°?

(a) $\vec{s} = (-2.00\hat{\imath} + 2.00\hat{\jmath})\,\mathrm{m}$, $s = 2.83\,\mathrm{m}$ at 45° north of west
(b) $F_x = 20.0\,\mathrm{N}$, $F_y = 34.6\,\mathrm{N}$:
$W = (20.0)(-2.00) + (34.6)(2.00) = 29.3\,\mathrm{J}$

Independent application

A rope force of $40.0\,\mathrm{N}$ acts at 60.0° north of east. The object moves $2.00\,\mathrm{m}$ north, then $2.00\,\mathrm{m}$ west.
(a) Find $\vec{s}$. (b) Find the work done by the force. (c) The force acts at $\vec{r} = 0.500\hat{\imath}\,\mathrm{m}$ from a pivot; find $\vec{\tau}$.
Verify (b) with $Fs\cos\phi$ and (c) with $rF\sin\phi$ and the right-hand rule.

Predict first

Sketch first. Which quadrant is $\vec{s}$ in? Is ϕ less than 90°?

(a) $\vec{s} = (-2.00\hat{\imath} + 2.00\hat{\jmath})\,\mathrm{m}$, $s = 2.83\,\mathrm{m}$ at 45° north of west
(b) $F_x = 20.0\,\mathrm{N}$, $F_y = 34.6\,\mathrm{N}$:
$W = (20.0)(-2.00) + (34.6)(2.00) = 29.3\,\mathrm{J}$

Check: $\phi = 135^\circ - 60^\circ = 75^\circ$;
$(40.0)(2.83)\cos 75^\circ = 29.3\,\mathrm{J}$ ✓
(c) $\tau_z = r_x F_y - r_y F_x = (0.500)(34.6) - 0 = 17.3$, so $\vec{\tau} = 17.3\,\hat{\boldsymbol{k}}\,\mathrm{N\,m}$
Check: $(0.500)(40.0)\sin 60^\circ = 17.3\,\mathrm{N\,m}$; counter-clockwise from above ✓

What you should remember: the chain, Contact 1 (part 1)

Concept	Limitation that motivates the next	Leads to
Problem-solving framework	needs quantities to work with: "5 m away" does not find a bag	**Magnitude**
Magnitude	every point on a $5\,\mathrm{m}$ circle fits	**Direction**
Direction	mass has none; some quantities need both	**Scalars, vectors**
Scalars, vectors	$400\,\mathrm{m}$ round a track: distance cannot say where you are	**Displacement**
Displacement	how to name, draw and compare it?	**Vocabulary**
Vocabulary	two displacements in a row: what is the net?	**Opening problem**
Opening problem	$1\,\mathrm{km} + 2\,\mathrm{km}$ as numbers ignores direction	**Graphical addition**
Graphical addition	how big is $\vec{R}$? Try one line first	**Parallel, antiparallel**

What you should remember: the chain, Contact 1 (part 2)

Concept	Limitation that motivates the next	Leads to
Parallel, antiparallel	off the line, $3\,\mathrm{m} + 4\,\mathrm{m} \neq 7\,\mathrm{m}$	**Magnitudes do not add**
Magnitudes do not add	three or more vectors?	**Associativity**
Associativity	undo an addition?	**Subtraction**
Subtraction	drawings are only as accurate as a ruler	90° **vectors**
90° **vectors**	why does the hypotenuse follow a rule?	**Pythagoras**
Pythagoras	solves the opening problem; at 60° it gives $3.61\,\mathrm{km}$, yet at least $4.00\,\mathrm{km}$ is needed	**Limitation, general triangles**
General triangles	four legs mean three triangles	**Components**

What you should remember: the chain, Contact 2

Concept	Limitation that motivates the next	Leads to
Why components	how much of $\vec{A}$ lies along each axis?	**Components**
Components	given A and θ, how to get the parts?	**Resolving**
Signs, quadrants	angles measured from other axes	**Resolving carefully**
Resolving	backwards: $\tan^{-1}$ gives the same value for opposite vectors	**Reconstructing**
Reconstructing	many vectors, each with components	**Adding vectors**
Adding vectors	displacements given as magnitudes and compass directions	**Multiple vectors**
Multiple vectors	writing components in words is long	**Unit vectors**
Unit vectors	a unit vector only points along an axis	**Unit vector of $\vec{A}$**
Unit vector of $\vec{A}$	all examples are planar	**3D vectors**
3D vectors	we can add and subtract, but not multiply	**Dot product**

What you should remember: the chain, Contact 3

Concept	Limitation that motivates the next	Leads to
Dot product	which angle is ϕ? Tail to tail, not the triangle angle γ	**The angle ϕ**
The angle ϕ	what does a dot product do for a force and a displacement?	**Work**
Work	the angle between vectors given by components	**Angle between vectors**
Angle between vectors	a perpendicular push turns a door most, but $\vec{r} \cdot \vec{F} = 0$	**Cross product**
Cross product	does the order matter?	**Right-hand rule, properties**
Cross product	a force applied at a distance from a pivot	**Torque**
Torque	problems that combine several tools	**Integrated applications**

THE SEVEN TOOLKITS

1	**Directions and angles**	directional language, standard angle, quadrant, reference angle, compass directions
2	**Vector triangles**	head to tail, internal angle, Pythagoras or sine and cosine rules, $\lvert A - B\rvert \le R \le A + B$
3	**Components**	choose axes $\rightarrow$ sketch $\rightarrow$ predict signs $\rightarrow$ resolve
4	**Reconstructing a vector**	magnitude; direction by quadrant, reference angle, placement, check
5	**Adding by components**	RESOLVE $\rightarrow$ COMBINE $\rightarrow$ RECONSTRUCT
6	**Dot product**	two forms, sign check, angle between vectors; preview: work
7	**Cross product**	magnitude, right-hand rule, unit vectors, components, perpendicularity check; preview: torque

Extension: direction angles in 3D are in the supplementary notes.